

## Tutorial-2

Let us design a DB for the following requirements:

- A highly popular 'McM Sangeet company' plans to organize a competition for searching the right talents to train them for creating music albums. The company advertises their recruitment requirements on different channels (both print and digital media) for inviting talents by submitting their personal information, prior experience, and a 2-5 minutes media file, which is of either an audio file (for songs) or a video (for songs and/or playing music instruments), to a given URL in the advertisement. The company writes in the advertisement that the shortlisted candidate will be informed by Phone and/or Email.
- A candidate can submit more than one entry and has an option to provide more than one phone number in the submission, which is unique to him/her.
- After the closing date of entry submission, a panel evaluates all entries and recommends a set of candidates to invite them for the next round. The company maintains the information about the panellist (personal information, industry experience, association with the McM company). A candidate can see the outcome of their application online.
- These shortlisted candidates are then invited to perform a 'Live' show in Mumbai and finally top ' $n$ ' candidates are selected in each album category (i.e. audio and video). These  $2n$  candidates are called as the members of McM-2020.
- Different music groups (pop, classic, leisure, evergreen, ...) are formed to create the music albums (audio/video). Every member belongs to one or more music groups which is moderated by a director who himself/herself is a member of the group. Each member has a different role to play in each album.
- Once the album is created and approved by McM Director, its trailer is released online for limited time to the outsiders to give their comments (like and dislike), and all this recorded in the database (album name, date of release, number of visits, number of likes/dislikes, ...).
- The McM decides the price of the album after analysing the data collected from its trailer release, and then the album is released to distributors who eventually will sell it online. Note, each distributor may be charged a different price per unit depending on the negotiation between the McM and distributor.
- When a distributor sells an album, the download request comes to the McM site. Thus, the McM company maintains the record for each download (Incoming URL – identifying the distributor, Album#, Date, Download Status – success/failure) so that they can track the number of downloads for raising the invoice.

Q1. Write the Relational Algebraic Expression (RAE) for each of the following query on relational schemas that was discussed in Tutorial-1.

A. List all 'Audio' albums released in 2020.

- B. List all participants who have submitted both Audio and Video files.
- C. List all members who have been the member of more than one group
- D. List all members of 'Pop' music group who are not part of any other music group.
- E. List all distributors who sold all types of albums.

Q2. Write the SQL statement on the same relational schemas for the following.

- A. List all 'Audio' albums released in 2020.
- B. List all members who have been the member of more than one group.
- C. List all members of 'Pop' music group who are not part of any other music group.
- D. List all participants who have submitted both Audio and Video files.
- E. The McM company would like to analyze the data for "Which advertisement channel has been effective in attracting the maximum number of entry submissions?"